

PROJECT SPECIFICATIONS

Sales and Marketing Agent Development for Online Website

1. User Specification

Parameter	Specification
User Type	Customers, Sales Executives, Marketing Managers, Business Owners, Administrators
Skill Level	Basic Computer Knowledge
Technical Expertise	No programming knowledge required
Internet Connection	Required for AI features, LLM APIs, CRM and cloud integration
Authentication	Email/Password, Google Login
Supported Devices	Desktop, Laptop, Tablet, Mobile
Supported Browsers	Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
Supported Languages	English, Hindi, Marathi (Extendable)
User Permissions	Chat with AI agents, submit inquiries, view recommendations, generate leads, access marketing content

2. System Specification

a. Hardware

Parameter	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or above
RAM	8 GB minimum (16 GB Recommended)
Storage	256 GB SSD (500 GB Recommended)
Graphics	Integrated Graphics (Dedicated GPU Optional)
Internet	Stable Broadband Connection
Display	1366 × 768 Resolution or Higher

b. Software

Parameter	Specification
Operating System	Windows 10/11 or Linux (Ubuntu)
Frontend	React.js, HTML5, CSS3, Tailwind CSS
Backend	FastAPI (Python)
Programming Language	Python 3.11+, JavaScript (ES6)
Database	PostgreSQL / MongoDB
AI Framework	LangChain, LangGraph / CrewAI
LLM	OpenAI GPT / Google Gemini / Llama
RAG	Retrieval-Augmented Generation
Vector Database	ChromaDB / Pinecone
ML Libraries	Scikit-learn, Pandas, NumPy
IDE	Visual Studio Code / PyCharm
API Testing	Postman
Version Control	Git & GitHub
Browser	Google Chrome / Microsoft Edge

3. Input Specification

Parameter	Specification
Input Type	Customer Query / Business Information
Customer Query	Natural Language Text
Lead Information	Name, Email, Company, Industry, Requirements, Budget, Timeline
Business Data	Products/Services, Target Industry, Customer Requirements
Knowledge Sources	Company Documents, FAQs, Product/Service Information
Marketing Input	Industry, Target Audience, Campaign Objective
Multiple Queries	Supported
User Interaction	Chat-based conversation with AI agents

4. Output Specification

Parameter	Specification
AI Chat Response	Context-aware answers to customer queries
Lead Qualification	Lead score and qualification status
Lead Summary	Automatically generated customer and requirement summary
Product/Service Recommendation	Personalized recommendations based on customer requirements
Marketing Content	AI-generated emails, promotional messages and campaign content
Sales Insights	Customer interests, requirements and conversion-related insights
RAG Response	Knowledge-grounded and context-aware answers
CRM Record	Customer details, conversation history and lead information
Agent Report	Summary of sales and marketing activities

5. Functional Specification

Parameter	Specification
Customer Interaction	AI-powered conversational interaction with website visitors
Sales Agent	Answers product/service-related questions and assists potential customers
Marketing Agent	Generates personalized marketing content and campaigns
Lead Qualification Agent	Analyzes customer information and automatically scores leads
RAG System	Retrieves relevant information from the organization's knowledge base
Multi-Agent Orchestration	Coordinates Sales, Marketing and Lead Qualification agents
Lead Management	Stores and manages qualified customer leads
Customer Profiling	Extracts industry, requirements, budget and purchase timeline
Personalization	Generates responses and recommendations based on customer profiles

Follow-up Automation	Generates appropriate follow-up messages for potential leads
Analytics	Provides lead, customer interaction and marketing insights
CRM Integration	Stores customer and lead information for sales management
Multi-language Support	English, Hindi, Marathi
Export Options	PDF, Excel, CSV, JSON

6. Non-Functional Specification

Parameter	Specification
Performance	Process customer queries and lead information efficiently
Response Time	AI responses should be generated within a few seconds
Security	JWT Authentication, Password Encryption and HTTPS
Availability	24 × 7 Web Access
Reliability	Accurate and consistent AI responses
Scalability	Supports multiple concurrent website users
Maintainability	Modular architecture for easy updates and agent integration
Compatibility	Windows, Linux, macOS and modern web browsers
Usability	Responsive and user-friendly interface
Data Integrity	Accurate storage and processing of customer and lead information
AI Safety	RAG-based responses to reduce inaccurate or unsupported information